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Physics-based Approximation and Prediction of Speedlines in Compressor Performance Maps

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Speedlines in compressor performance maps (CPMs) are critical for understanding and predicting compressor behavior under various operating conditions. We investigate a physics-based method for reconstructing compressor performance maps from sparse measurements by fitting each speedline with a superellipse and encoding it as a compact, interpretable vector (surge, choke, curvature, and shape parameters). Building on the formulation of Llamas et al., we develop a robust two-stage fitting pipeline that couples global search with local refinement. The approach is validated on industrial data-sets for different turbocharger types. We discuss prediction quality for inter- and extrapolation, metric sensitivities and outline opportunities for physics-informed constraints, alternative function families, and hybrid physics-ML mappings to improve boundary behavior and, ultimately, enable full CPM reconstruction from limited data.

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